

T1D Field Management and Insulin Pen Survival Guide (v2.0)

An updated, evidence-grounded field guide for Type 1 Diabetes management in austere, off-grid, or combat conditions. Refined based on Wilderness Medical Society Clinical Practice Guidelines (2019) and operational medical peer review (2025). This version integrates insulin-type stability data, altitude adjustments, illness rules, and power management refinements.

1. Temperature Limits and Storage

Maintain insulin between 2–30 °C (36–86 °F). Avoid freezing or direct sunlight. Potency loss increases above 37 °C or after freeze–thaw. Use evaporative cooling pouches (e.g., FRÍO) for hot environments. In cold, keep pens near the body core. Never place insulin directly on ice or heat sources. Rotate older pens first; discard if cloudy or crystallized. Open pens remain viable ~28 days below 30 °C. Insulin Type Heat Tolerance: • Rapid-acting (Lispro, Aspart): stable up to 30 °C (~86 °F) for 28 days. • Long-acting (Glargine, Degludec): slightly more heat-resistant, tolerating brief exposure up to 31–32 °C before notable potency loss. In extreme heat, use newer pens first and rotate rapid-acting stock before basal reserves.

2. Heat vs. Cold Resilience

Pens have low thermal mass, making them more reactive to temperature swings than vials. Vials resist heat and cold slightly better, but pens are more rugged and convenient. Pens heat faster in the sun but also cool faster in evaporative wraps. In freezing conditions, keep pens insulated and close to body warmth; frozen insulin must be discarded immediately.

3. Monitoring Protocols and Power Discipline

Without CGM, use finger-stick testing: • Before exertion, every 1–2 hours during activity, and at any symptom (confusion, shakiness, fatigue). • Camp check at bedtime and again around 0200 hrs to catch delayed hypoglycemia. Carry 2 meters, extra batteries, and at least 600 strips for one month (15/day + buffer). Check ketones if BG > 250 mg/dL, or during illness or vomiting. Power Supply Discipline: Use lithium (CR2032 or AAA) cells—cold-resistant and 5-year shelf life. Store spares vacuum-sealed with desiccant and check voltage monthly.

4. Hypoglycemia (Low Blood Glucose <70 mg/dL)

Symptoms: shaking, sweating, confusion, tachycardia. Field treatment: 15–20 g fast carbohydrate → wait 15 min → recheck → repeat until >90 mg/dL. Follow with slower carbs. If unconscious, administer glucagon and place in recovery position. Evacuate if unresponsive after 5 minutes. Glucagon for Field Use: Prefer prefilled nasal or autoinjector forms (Baqsimi, Gvoke) for speed and reliability under stress. Powder-mix kits are difficult in darkness or cold. Shelf life ≈2 years; store near body warmth in heat or cold extremes.

5. Hyperglycemia and Ketones

BG > 250 mg/dL: Check ketones. • Trace/none: Hydrate (250 mL every 30 min), small correction dose, recheck in 1–2 h. • Moderate/high or vomiting: Stop exertion, hydrate aggressively (500 mL every 30 min), correction insulin, rest. • Evacuate if vomiting persists or mental status declines (DKA risk).

6. Illness or Vomiting Protocol (Sick-Day Rule)

When sick or vomiting: • Check BG and ketones every 2–3 h. • If BG > 250 mg/dL without ketones, increase basal 10–20%. • Maintain hydration 250–500 mL/hr. Evacuate if unable to retain fluids, ketones remain moderate/high >6 h, or symptoms worsen.

7. Hydration and Electrolyte Management

Hydration prevents DKA and moderates BG swings. Maintain clear/pale urine. Replace electrolytes every few hours in heat or high exertion. Target 0.5–1 L/hour during activity in heat. Use water + electrolyte tablets or powders. Avoid alcohol and high-caffeine energy drinks.

8. Environmental Effects on Insulin and Physiology

Heat increases absorption → risk of hypoglycemia. Cold slows absorption → risk of hyperglycemia. Altitude (>4000 m): adrenaline and cortisol rise, increasing insulin resistance. Expect mild BG rise; if trend persists >24 h, increase basal/correction ~10%. Hydration and rest help offset hormonal stress. Infection or high stress increases BG and insulin needs. Rotate injection sites regularly to reduce absorption variability.

9. Storage and Packing Strategy (Pen-Focused)

Carry configuration: • Active pen (daily use): inner pocket or thermal pouch near body temperature. • Backup pens: sealed in evaporative or insulated wrap deep in pack. • Redundancy: one active + one spare per insulin type per 2 weeks. A one-month supply for 40 U/day = ~4–5 pens (~60 g). Add buffer pens and one unused “cold reserve.”

10. Injection Hygiene and Safety

Use new needles whenever possible; if reusing, cap immediately and limit to ≤3 times. Clean site with alcohol or clean water + friction if needed. Dispose of sharps in rigid plastic bottle or secure container. Maintain hand hygiene before dosing. Rotate injection sites for consistent absorption in varying temperatures.

11. Field Decision and Evacuation Triggers

Evacuate if: • Persistent vomiting, confusion, or rapid breathing (DKA suspicion). • BG > 300 mg/dL + moderate/high ketones not improving after 2 hours. • Loss of insulin supply or meter function >24 h. • Severe hypo not recovering after glucagon. Stay and manage only if stable and responsive.

12. Core Field Principles

1. Redundancy beats technology: duplicate all critical gear. 2. Manual logs outperform electronics when power fails. 3. Hydration is a life-saving priority. 4. Partner training in glucagon and recognition saves lives. 5. Aim for safe, not perfect, glucose: 90–250 mg/dL during field operations. 6. Rotate supplies and protect insulin from light, heat, and freezing.

Summary of Evidence Alignment

This v2.0 revision integrates refinements suggested by field-medicine reviewers, including insulin chemistry nuances, altitude effects, sick-day rules, and power management. It remains directly aligned with the Wilderness Medical Society 2019 Diabetes Guidelines and reflects operational doctrine for austere, no-tech survival scenarios.